

G. W. Leibniz
ANALYSIS GEOMETRICA PROPRIA

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Gerhardt (ed.), *Leibnizens mathematische Schriften*, Bd. V (1858), S. 172–178 (Stück II).

5 *Analysin Geometricam propriam* eique connexum *calculus*
situs, hic attentabimus nonnihil speciminis gratia, ne forte
pereat cogitatio, quae aliis quod sciam in mentem non venit,
et usus longe alios ab iis dabit, quos Algebra praestat. Scien-
10 dum enim est diversae considerationis esse magnitudinem et
situm. Magnitudo datur etiam in illis quae situm non habent,
ut numerus, proportio, tempus, velocitas, ubicunque scilicet
partes existunt, quarum numero seu repetitione aestimatio
fieri potest. Itaque eadem est doctrina magnitudinis et nu-
15 merorum, Algebrae ipsa vel si mavis Logistica, tractans de
magnitudine in universum, revera agit de numero incerto vel
saltem innominato. Sed situs magnitudini vel partium multi-
tudini formam quandam superaddit, ut in numeris figuratis.
Hinc patet, Algebram continere Analysin proprie et per se
15 ad Arithmeticam pertinentem, etsi ad Geometriam et situm
transferatur, quatenus linearum et figurarum magnitudines
tractantur. Sed tunc necesse est, Algebram multa supponere
propria Geometriae vel situs, quae et ipsa resolvi debebant.
Analysis igitur nostra resolutionem illam effectui dat nihil
20 amplius assumens aut supponens nec tam magnitudini, quam
ipsi per se situi accommodata. Nunc autem ad explicandam
rem situs non nisi *congruentia* utemur, sepositis in alium
locum *similitudine* et *motu*.

(1) *Congrua* sunt quae sibi substitui possunt in eodem loco,

5R We will attempt here, as something of a sample, *properly geometric*
analysis and the *calculus of situation* connected with it, lest by chance
the thought is lost, which has not come to mind for anyone else, as
far as I know, and which will provide very different applications from
those furnished by algebra. What needs to be understood is that mag-
10R nitude and situation are different considerations. There is magnitude
even in things that do not have situation, such as number, proportion,
time, velocity; namely wherever there exist parts which can be as-
sessed through number, i.e. by repetition. Therefore the doctrines of
magnitudes and of numbers are the same; and algebra itself, or logistic
(if you prefer), when treating of magnitude in general, in fact deals
with unspecified, or at least unnamed, number. But situation adds a
certain form over and above the magnitude or the multitude of parts,
as in figurate numbers. Hence, algebra clearly comprises analysis
15R properly and in itself belonging to arithmetic, even if it is transferred
to geometry and situation to the extent that it handles the magnitudes
of curves and figures. But then algebra necessarily supposes much
that is proper to geometry or situation which should be released from
it. Our analysis, therefore, effects this release, assuming or supposing
20R nothing more, and being not so much accommodated to magnitude as
to situation itself. Moreover, for explaining the matter of situation we
now use nothing but *congruence*, reserving *similarity* and *motion* to
another place.

(1) Things are *Congruent* which can be substituted for one another

ut ABC et CDA (fig. 65), quod sic designo $ABC \simeq CDA$. Nam $\sim$ mihi est signum similitudinis, et $=$ aequalitatis, unde congruentiae signum compono, quia quae simul et similia et aequalia sunt, ea congrua sunt.

in the same place, as ABC and CDA [Figure 1], which I signify thus: $ABC \simeq CDA$. For $\sim$ is my sign of similarity, and $=$ of equality, from which I compose the sign of congruence, since things that are simultaneously similar and equal are congruent.

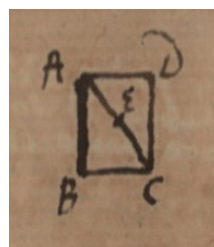


Figure 1

5 (2) *Eodem modo se habere* hic censentur vel *congruenter*, ea quae in congruis sibi respondent. Exempli causa $AB.ABC \simeq CD.CDA$, nempe AB est ad ABC , ut CD ad CDA ; ita et $AB.AC \simeq CD.CA$ et $A.BC \simeq C.DA$, id est punctum A eodem modo se habet ad rectam BC , ut punctum C ad DA . Neque
10 enim hic tantum de ratione aut proportionione agitur, sed de relatione quacunque.

(3) *Axioma*: Si determinantia sint congrua, talia erunt etiam determinata, posito scilicet eodem determinandi modo. Exempli gratia si $A.B.C \simeq D.E.F$, etiam circumferentia circuli per $A.B.C$ congruet circumferentiae circuli per $D.E.F$, quia
15 datis tribus punctis circumferentia circuli est determinata. Et in universum in calculo congruentiarum substitui possunt determinata pro determinantibus sufficientibus, prorsus ut in vulgari calculo aequalia aequalibus substituuntur. Vid.
20 infra § 26. § 30. § 32.

(4) Ut autem calculus melius intelligatur, notandum est, cum dicitur $A.B.C \simeq D.E.F$, idem esse ac si simul diceretur $A.B \simeq D.E$. et $B.C \simeq E.F$ et $A.C \simeq D.F$, ita ut haec ex illo fieri

5R (2) Things are counted as *relating in the same manner*, or *congruently*, which correspond to each other in congruent things. For example, $AB.ABC \simeq CD.CDA$, namely AB is to ABC as CD to CDA , and so $AB.AC \simeq CD.CA$ and $A.BC \simeq C.DA$, that is, the point A relates in the same manner to the line BC as the point C to DA . Indeed here it
10R is done not just by ratio or proportion, but by whatever relation.

(3) *Axiom*: If the determiners are congruent, so too will be the things determined, supposing of course the same way of determining. For example, if $A.B.C \simeq D.E.F$, then the circumference of the circle through $A.B.C$ is also congruent to the circumference of the circle through
15R $D.E.F$, since the circumference of a circle is determined by three given points. And in general, in the calculation of congruents, determined things may be substituted for sufficient determiners, just as equals are substituted for equals in common calculation. See below §26, §30, §32.

20R (4) In order to understand the calculus better, it should be observed that when we say $A.B.C \simeq D.E.F$, it is the same as if we had said simultaneously: $A.B \simeq D.E$ and $B.C \simeq E.F$ and $A.C \simeq D.F$, so that

possint *divellendo*, et illud ex his *conjungendo*. Vid. infra § 26. 27. § 29. § 30. § 31. § 32.

(5) *Terminus communis* est locus qui inest duobus locis, ita ut pars eorum non sit. Sic punctum *E* est locus, qui inest rectis *AE*, *CE*, pars autem est neutrius. Itaque terminus earum communis esse dicitur.

(6) *Sectio* est duarum partium totum constituentium nec partem communem habentium terminus communis totus. Ita *AC* est terminus communis totus triangulorum *ABC*, *CDA* constituentium totum *ABCD* nec habentium partem communem.

(7) Hoc ita calculo exprimemus, per quem Geometria ad Logicam refertur. Punctum omne in proposito loco existens communi nota vel litera designetur (exempli gratia) *X*, et ipse locus designetur per $\overline{X}$, lineam super litera ducendo. Si quavis loci puncta sint *Y* et *Z*, loca erunt $\overline{Y}$ vel $\overline{Z}$. Sit ergo totum $\overline{X}$, partes constituentes sint $\overline{Y}$ et $\overline{Z}$, et sectio sit $\overline{V}$, formari poterunt hae propositiones: Omne *Y* est *X*, omne *Z* est *X*, quia $\overline{Y}$ et $\overline{Z}$ insunt ipsi $\overline{X}$. Sed et quod non est *Y* nec *Z*, id non est *X*, posito $\overline{Y}$ et $\overline{Z}$ esse partes constituentes seu exhaustientes totum $\overline{X}$. Porro omne *V* est *Y*, et omne *V* est *Z*, quia $\overline{V}$ est ipsis $\overline{Y}$ et $\overline{Z}$ commune, seu utrique inest. Denique quod est *Y* et *Z* simul, id etiam est *V*, quia $\overline{V}$ est sectio seu terminus communis totus, scilicet qui continet quicquid utrique commune est, partem enim (seu aliquid praeter terminum) non habent communem. Hinc omnes Logicae subalternationes, conversiones, oppositiones et consequentiae hic locum interdum cum fructu habent, cum alias a realibus proscriptae fuerint visae, hominum vitio, non propria culpa.

these can be made from that by *disjoining*, and that from these by *combining*. See below §26, §27, §29, §30, §31, §32.

(5) A *common boundary* [*terminus*] is a locus which is in [*inest*¹] two loci, such that it is not a part of them. In this way a point *E* is a locus which is in the lines *AE*, *CE*, but is not part of either. Therefore it is said to be their common boundary.

(6) A *section* is the entire common boundary of two parts constituting a whole and not having a common part. Thus *AC* is the entire common boundary of triangles *ABC*, *CDA* constituting the whole *ABCD* and not having a common part.

(7) We will express this by the calculus, through which geometry is carried back to logic, in this way: Every point that exists in a proposed locus is designated by a common mark or letter (for example) *X*, and the locus itself is designated by $\overline{X}$, by drawing a line over the letter. If some given points of the locus are *Y* and *Z*, [their] loca will be $\overline{Y}$ or $\overline{Z}$. Let, therefore, the whole be $\overline{X}$, the constituting parts be $\overline{Y}$ and $\overline{Z}$, and $\overline{V}$ be the section; then these propositions can be formed: Every *Y* is *X*, every *Z* is *X*, since $\overline{Y}$ and $\overline{Z}$ are in $\overline{X}$. But also, whatever is neither *Y* nor *Z* is not *X*, having set $\overline{Y}$ and $\overline{Z}$ to be *parts constituting*, or *exhausting*, the whole $\overline{X}$. Further, every *V* is *Y*, and every *V* is *Z*, since $\overline{V}$ is in *common* to $\overline{Y}$ and $\overline{Z}$, i.e. is in both of them. Finally, whatever is simultaneously *Y* and *Z* is also *V*, since $\overline{V}$ is the *section*, i.e. the entire common boundary, namely that which contains what is common to both, and indeed they don't have a common part (i.e. something besides a boundary). Hence all subalterns, conversions, oppositions, and consequences of logic have a place here, sometimes fruitfully, although other times they seem to be precluded from real [use] by human weakness, not their own fault.

¹Leibniz is probably using *inesse* in his technical sense associated with his mereological calculus: *X inest Y* when, *X* being supposed, *Y* is immediately thereby also supposed. We do not flag subsequent instances in this essay.

(8) *Coincident* loca $\bar{X}$ et $\bar{Y}$, si omne X sit Y , et omne Y sit X . Hoc ita designo: $\bar{X} \approx \bar{Y}$.

(8) The loci $\bar{X}$ and $\bar{Y}$ *coincide* if every X is Y and every Y is X . I signify it thus: $\bar{X} \approx \bar{Y}$.

(9) *Punctum* est locus, in quo nullus alius locus assumi potest, itaque si in puncto $\odot$ assumatur locus $\triangleright$, coincidet $\triangleright$ ipsi $\odot$, et vicissim si $\triangleright$ insit in $\odot$ et ex hoc solo concludatur coincidere $\odot$ et $\triangleright$, erit $\odot$ punctum.

(9) A *point* is a locus in which no other locus can be taken; and so if the locus $\triangleright$ is taken in the point $\odot$, then $\triangleright$ coincides with $\odot$, and if in turn $\triangleright$ is in $\odot$ and from this alone one may conclude that $\odot$ and $\triangleright$ coincide, then $\odot$ is a point.

Spatium absolutum est contrarium puncto; nam in spatio omnis alius locus assumi potest, ut in puncto nullus, ut ita punctum sit simplicissimum in situ, et velut minimum, spatium vero sit diffusissimum et velut maximum.

Absolute space is the opposite of a point, for in space every other locus can be taken, as none can be taken in a point, so the point is the simplest with respect to situation, and as it were the minimum, whereas space is the most diffuse, and as it were the maximum.

(10) *Corpus* (mathematicum scilicet) seu *solidum* est locus, in quo plus est quam terminus. Atque hoc scilicet volumus, cum solido tribuimus profunditatem. Contra quicquid in superficie aut linea est, terminus intelligi potest, et commune esse alicui cum alio partem communem cum ipso non habente. Analogia hic etiam est inter punctum et solidum, quod quicquid puncto inest, punctum est; contra cui solidum inest, id solidum est. Item, punctum non potest cuiquam inesse ut pars; at solidum nulli aliter inesse potest quam ut pars.

(10) A *body* (mathematical, of course), or *solid*, is a locus in which there is more than boundaries. And this is what we mean when we attribute depth to a solid. Conversely, whatever is in a surface or curve can be understood to be a boundary, and is in common to something together with another thing not having a common part with it. There is an analogy between point and solid here also, that whatever is in a point is a point, and conversely whatever a solid is in, is a solid. Likewise, a point cannot be in something as a part; but a solid cannot be in anything other than as a part.

(11) *Planum* est sectio solidi utrinque eodem modo se habens ad ea, quae solidi terminos non attingunt, seu utrinque eodem modo se habens ad ea, quae fiunt in una parte ut in alia. Si pomum plano secas, duorum segmentorum extrema, quibus cohaerebant, distingui invicem non possunt.

(11) A *plane* is a section of a solid relating in the same way on both sides² to what does not touch the boundaries of the solid, i.e. relating in the same way on both sides to what ends up on one side as to what ends up on the other. If you cut an apple by a plane, the faces [*extrema*] of the two pieces, by which they were attached, cannot be distinguished from each other.

(12) Itaque si solidum interminatum sit, absolute verum est, planum secans utrinque eodem modo habere. Sin terminatum sit solidum, sufficit terminos in rationes non venire. Et utrinque eodem modo facta etiam ad sectionem se eodem

(12) And so if it is an unbounded solid, it is absolutely true that the dividing plane relates in the same way to both sides. But if the solid is bounded, it is enough that the boundaries do not enter into consideration. And something done the same way on both sides will

²'Relating in the same way' is our fixed expression for *eodem modo se habens*. Here and below it is joined with *utrinque*, 'on or from both sides or parts'.

modo habebunt.

(13) *Recta* est sectio plani utrinque eodem modo se habens ad ea, quae terminos plani non attingunt.

(14) Sit planum interminatum *AA* (fig. 66), ejusque sectio *BB*
5 utrinque se habens eodem modo, erit *BB* recta interminata.

indeed relate in the same way to the section.

(13) A *line* is a section of a plane relating in the same way on both sides to what does not touch the boundaries of the plane.³

(14) Let there be an unbounded plane *AA* [Figure 2], and its section *BB*
5R relates in the same way on both sides; then *BB* will be an unbounded line.



Figure 2

(15) Sed et si planum sit terminatum *CC* (fig. 67), quaecunque ejus figura sit, si tamen tegamus terminos ut non appareant, vel rationem eorum nullam habeamus, reperiemus rectam secantem *DD* utrinque se eodem modo habere, eaque erit
10 terminata.

(15) But if a plane *CC* is bounded [Figure 3], no matter what its shape, if we nonetheless cover the boundaries so that they don't appear, or we in no way take them into account, then we will discover a dividing line *DD* to relate in the same way on both sides, and it will be bounded.

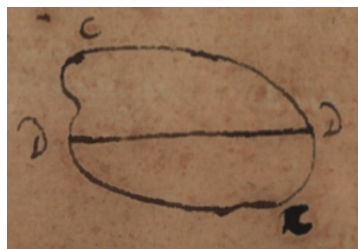


Figure 3

³Leibniz added the following marginal remark: 'What if someone doubted whether the plane could be so divided? Is it preferable, therefore, to form a line as the intersection of two planes?'

(16) *Curva* vero diversimode se habet utrinque, cum ab uno latere sit concava, ab alio convexa.

(16) But a curve relates differently on each side, since it is concave on one side, and convex on the other.

Quae sequentur, nunc quidem omnia in plano intelligantur.

Now, everything that follows should be understood to be in a plane.⁴

(17) Si sit recta (fig. 68), in qua puncta *A* et *B*, et extra eam punctum *C* ab uno latere, tunc oportet dari posse aliud *D* ab altero latere, quod eodem modo se habeat ad *A* et *B*, quo *C* se ad ea habet. Nam alioqui cum puncta haec sint in recta ex hypothesi, unum latus non ita ut alterum ad rectam se haberet, contra rectae definitionem. — Dato igitur *C.A.B* inveniri potest *D*, ut sit *C.A.B* $\simeq$ *D.A.B*.

(17) If there is a line [Figure 4] in which there are points *A* and *B*, and outside it a point *C* on one side, then necessarily another *D* can be given on the other side, which relates in the same way to *A* and *B* as *C* does to them. For otherwise, since these points are in the line by hypothesis, one side would not relate to the line like the other, contrary to the definition of a line. Therefore, given *C.A.B*, one can find *D* such that *C.A.B* $\simeq$ *D.A.B*.

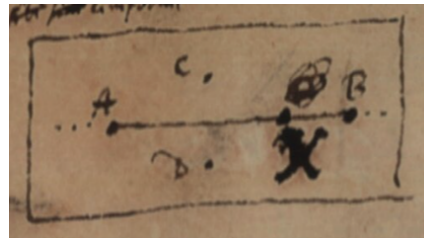


Figure 4

(18) Itaque si detur punctum *X* suae ad duo puncta *A.B* relationis unicum, id non poterit esse ab alterutro latere rectae per *A.B*; alioqui contra hypothesin daretur aliud ei geminum, per praecedentem. — Hinc necesse est ut cadat in ipsam rectam, ubi gemina alibi in unum coeunt, cum recta sit utriusque lateris terminus communis.

(18) Therefore if a point *X* is given, unique with its relation to two points *A.B*, it could not be on either side of the line through *A.B*, otherwise another one would be given by the preceding, its twin, contrary to hypothesis. Hence it is necessary that it falls on that very line, where what are elsewhere twins merge into one, since the line is also the common boundary of both sides.⁵

⁴Leibniz's marginal note: 'It seems necessary for some property of the plane to enter the argument, such as that two planar figures with the same boundary are congruent, i.e. that the interior is uniform.'

⁵Leibniz's marginal note: 'The converse still needs to be demonstrated, that every point in a line is unique with its relation to two points in it.'

In turn, every point in the line through *A.B* is unique with its relation to those two points. Let such a point be *X*; if it is not unique, then an Ω is given which is to *A.B* as *X* is to *A.B*; since *X* is in the line through *A.B* by hypothesis, then so too is Ω ; there is therefore some order among the four points *A.B.X* Ω in the straight line, therefore *A.B.X* and

(19) Recta igitur (terminata scilicet) est locus omnium punctorum suae ad duo in ipsa puncta relationis unicum. Sit $X.A.B \simeq Z.A.B$, atque ideo X coincidat ipsi Z , erit $\bar{X}$ recta (interminata) per $A.B$.

5 (20) Unde jam colligitur, duas rectas non posse sibi occurrere nisi in uno puncto, seu duas rectas, quae habeant duo puncta A et B communia, productas coincidere inter se, cum utraque sit locus omnium (atque adeo eorundem) punctorum suae ad puncta A et B relationis unicum. Atque ita datis duobus
10 punctis determinata est recta, in quam cadunt.

(21) Hinc porro duae rectae, quae scilicet productae non coincidunt, non possunt habere segmentum commune. Nam si segmentum commune AB (fig. 69) habeant, duo etiam puncta minimum habebunt communia $A.B$, ergo productae
15 coincident.

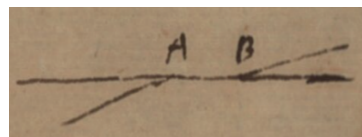


Figure 5

(22) Similiter duae rectae non possunt claudere spatium, alioqui bis sibi occurrerent, adeoque duo puncta communia haberent (fig. 70).

(19) Consequently, a straight line (bounded, of course) is the locus of all points unique with their relation to two points in itself. If $X.A.B \simeq Z.A.B$, and for that reason X coincides with Z , then $\bar{X}$ is the (unbounded) line through $A.B$.⁶

5R (20) From this one now concludes that two lines could not meet each other except in one point, i.e. two lines which have two points A and B in common coincide with each other when extended, since either one is the locus of all points (and thus of the same ones) unique with their relation to points A and B . And so, given two points, a line is
10R determined on which they fall.

(21) Hence, moreover, two lines, namely which do not coincide when extended, cannot have a segment in common. For if they have a segment AB in common [Figure 5], then they also have at least the two points $A.B$ common; therefore [when] extended they coincide.

15R (22) Similarly, two lines cannot enclose space, otherwise they will meet each other twice, and thus they will have two common points [Figure 6].

$A.B.\Omega$ are not $\simeq$. Supposing that the locus is a *curve*, there is not an order of all points in the (linear?) curve. Therefore there are two points in it relating in the same way to two points in itself, and so the locus determined with respect to the two is a curve.'

⁶Leibniz's marginal note: 'In general, every point in a curve not returning to itself is unique with its relation to two points taken in it.'



Figure 6

(23) Ita ex nostra rectae definitione demonstrantur Axiomata, quae Euclides sine demonstratione circa rectam assumsit.

(23) Thus, from our definition of the line we demonstrate axioms which Euclid assumes about the line without demonstration.

(24) *Circulus* fit rectae circa unum punctum quiescens motu in plano. Extremum quiescens erit centrum, linea ab altero extremo descripta erit circumferentia.

(24) A *circle* is produced by the motion of a line around a single point remaining stationary in the plane. The stationary extremum is the centre, and the curve described by the other extremum is the circumference.

(25) Itaque (fig. 71) omnia circumferentiae puncta, ut X , sese eodem modo habebunt ad centrum C , seu omnia X se habebunt ad C , ut A ad C . Quod ut calculo nostro exprimatur, si sit $X.C \simeq A.C$, erit $\bar{X}$ *circuli circumferentia*.

(25) And so [Figure 7] every point of the circumference, say X , will relate in the same way to the centre C , i.e. every X will relate to C as A does to C . As expressed in our calculus, this is: if $X.C \simeq A.C$, then $\bar{X}$ will be the *circumference* of the *circle*.

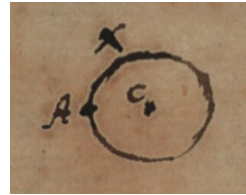


Figure 7

(26) Locus omnium punctorum eodem modo se ad duo puncta habentium est *recta*. Sint duo puncta C et D (fig. 72), sitque locus $\bar{X}$, cujus quodlibet punctum X eodem modo se habeat ad C quo ad D ; dico $\bar{X}$ esse *rectam*. — Quod ut demonstretur, in loco $\bar{X}$ sumantur duo puncta A et B , ducatur *recta* $\bar{Z}$ per A et B ; ea determinata est ex ipsis $A.B$ per § 20. Jam $A.B.C \simeq A.B.D$ ex hypothesi, quia A et B cadunt in $\bar{X}$; ergo (per axioma § 3) etiam *recta* per $A.B$ seu $\bar{Z}$ eodem modo se habebit ad C ut ad D ,

(26) The locus of all points relating in the same way to two points is a *straight line*. Let there be two points C and D [Figure 8], and let there be a locus $\bar{X}$ any point X of which relates in the same way to C as to D ; I claim that $\bar{X}$ is a line. In order to demonstrate this, let two points A and B be taken in the locus $\bar{X}$, and the line $\bar{Z}$ be drawn through A and B ; it is determined from $A.B$ by §20. Now $A.B.C \simeq A.B.D$ by hypothesis, since A and B fall in $\bar{X}$; therefore (by an axiom in §3) the line through $A.B$, or $\bar{Z}$, will also relate in the same way to C as to D ,

seu erit $\bar{Z}.C \simeq \bar{Z}.D$; jam etiam $X.C \simeq X.D$ ex hypothesi, ergo
 conjungendo $X.\bar{Z}.C \simeq X.\bar{Z}.D$. Ergo punctum X non potest
 esse ab uno latere rectae z , veluti (si placet) a latere D ; ita
 enim se aliter haberet ad rectam $\bar{Z}$ et ad D , quam ad rectam
 5 $\bar{Z}$ et ad C , itaque necesse est X cadere in $\bar{Z}$ seu omne X erit
 Z , unde et $\bar{X}$ cadet in $\bar{Z}$, quod erat demonstrandum.

or $\bar{Z}.C \simeq \bar{Z}.D$; and now too $X.C \simeq X.D$ by hypothesis, therefore by
 combining, $X.\bar{Z}.C \simeq X.\bar{Z}.D$. Therefore the point X cannot be on one
 side of the line $\bar{Z}$, such as (if you will) on the D side; for then it would
 relate differently to the line $\bar{Z}$ and to D than it does to the line $\bar{Z}$ and
 to C ; and so it is necessary that X falls in $\bar{Z}$, i.e. every X will be Z ,
 whence also $\bar{X}$ falls in $\bar{Z}$, *quod erat demonstrandum*.

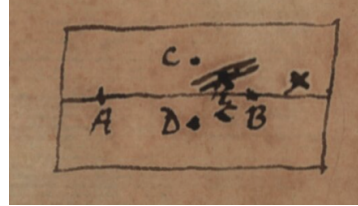


Figure 8

(27) Hic ergo specimen calculi habuimus non inelegans ad
 praescriptum § 4. Nempe quia $X.\bar{Z} \simeq X.\bar{Z}$, quod est iden-
 ticum, et $X.C \simeq X.D$ ex hypothesi, et $\bar{Z}.C \simeq \bar{Z}.D$, quod proba-
 10 tum dedimus ex natura rectae, ex his binionibus omnibus
 singulatim respective congruentibus sequitur congruere et
 conflatas inde terniones seu conjungendo esse

$$\underline{X.\bar{Z}.C} \simeq \underline{X.\bar{Z}.D}.$$

(28) Hinc si $X.C \simeq X.D$, erit $\bar{X}$ recta, quae congruentia per-
 15 magnae est utilitatis in calculo nostro. Et vicissim si $\bar{X}$ sit
 recta, oportet existere puncta qualia C et D , ut locum habeat
 congruentia.

(29) Fieri nequit, ut recta eodem modo se habeat ad tria plani
 puncta seu ut sit $X.C \simeq X.D \simeq X.E$ (fig. 73). Nam si hoc
 20 esset, foret conjungendo $X.C.E \simeq X.D.E$, ergo E non potest
 esse ab alterutro latere. Sed idem E non potest esse in $\bar{X}$, ita
 enim etiam C et D forent in $\bar{X}$, et hoc amplius, coinciderent cum
 E , alioqui punctum aliquod rectae (nempe ipsum E) se aliter
 haberet ad C et ad D quam ad E , ergo punctum E praeter C

(27) So we have here an elegant sample of the calculus following the
 rule of §4. Namely, since $X.\bar{Z} \simeq X.\bar{Z}$, which is identical, and $X.C \simeq$
 $X.D$ by hypothesis, and $\bar{Z}.C \simeq \bar{Z}.D$, which we proved from the nature
 10R of the line, from all these congruent pairs [*binionibus*] considered
 individually, it follows that the triples assembled from them are also
 congruent, i.e. that by combining,

$$\underline{X.\bar{Z}.C} \simeq \underline{X.\bar{Z}.D}$$

(28) Hence if $X.C \simeq X.D$, then $\bar{X}$ will be a *straight line*, which congru-
 15R ence is of the utmost utility in our calculus. And conversely, if $\bar{X}$ is
 a straight line, there should be points of such kind as C and D so a
 congruence has the locus.

(29) It cannot happen that a line relates in the same way to three
 points of the plane, i.e. that $X.C \simeq X.D \simeq X.E$ [Figure 9]. For if this
 20R were so, then by combining, $X.C.E \simeq X.D.E$, therefore E cannot be on
 either side. But the same E cannot be in $\bar{X}$, for then C and D would
 also be in $\bar{X}$, and moreover, they would coincide with E , otherwise
 some point of the line (namely E itself) would relate differently to C
 and to D than to E , therefore a point E in addition to C and D cannot

et D nuspiam reperiri potest.

be found anywhere.

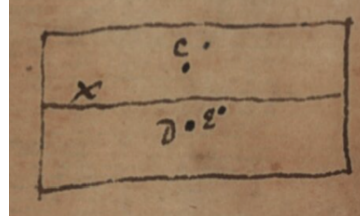


Figure 9

(30) Circulus circulo non occurrit in pluribus quam duobus punctis. Sint duae circumferentiae circulares $\bar{X}$ et $\bar{Z}$ (fig. 74), dico eas non posse secare nisi in duobus punctis, velut L et M . Nempe ipsius $\bar{X}$ centrum sit A , ipsius $\bar{Z}$ centrum sit B . Quia jam L est X et M est X , erit $LA \simeq MA$, et quia L est Z et M est Z , erit $LB \simeq MB$, utrumque ex natura circumferentiarum, quibus puncta sunt communia per § 25. Ergo conjungendo $LA.B \simeq MA.B$. Sit $\bar{Y}$ recta per $A.B$, utique etiam $L.\bar{Y} \simeq M.\bar{Y}$ per axiom. § 3; sed si daretur praeterea aliud duabus circumferentiis commune punctum N , haberemus $L.\bar{Y} \simeq M.\bar{Y} \simeq N.\bar{Y}$, seu recta $\bar{Y}$ se eodem modo haberet ad tria puncta L, M, N , quod fieri nequit per praecedentem. Hinc sequitur, tribus datis punctis circumferentiam, cui insint, esse determinatam, cum pluribus simul inesse non possint.

(30) A circle does not meet a circle in more than two points. Let two circular circumferences be $\bar{X}$ and $\bar{Z}$ [Figure 10]; I say they cannot cross except in two points, such as L and M . Indeed, let the centre of $\bar{X}$ be A , the centre of $\bar{Z}$ be B . Now since L is X and M is X , $LA \simeq MA$, and since L is Z and M is Z , $LB \simeq MB$, both from the nature of the circumferences to which the points are common by §25. Therefore, $LA.B \simeq MA.B$ by combining. Suppose $\bar{Y}$ is the line through $A.B$, so that now $L.\bar{Y} \simeq M.\bar{Y}$ by the axiom of §3; but if, in addition, another point N is given, common to the two circumferences, we would have $L.\bar{Y} \simeq M.\bar{Y} \simeq N.\bar{Y}$, or the line $\bar{Y}$ would relate in the same way to the three points L, M, N , which cannot happen by the preceding. Hence it follows that when three points are given, a circumference is determined which they are in [*insint*], since they cannot simultaneously be in multiple [circumferences].

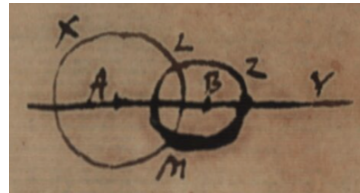


Figure 10

(31) Si circulus circulum tangat (fig. 75), punctum contactus

(31) If a circle is tangent to a circle [Figure 11], the point of contact is

in eadem est recta cum centris. Centra sunt A et B , et punctum contactus C , ubi scilicet duo puncta occursus coalescunt. Itaque (per praecedentem) circulus circulo praeterea non occurrit, alioqui forent puncta occursus tria. Punctum ergo
 5 contactus duobus circulis solum commune est. Ajo, id esse in recta per $A.B$. Quod patebit per § 19, si ostendatur, esse suae ad $A.B$ relationis unicum. Esto aliud, si fieri potest, F et debet esse $F.A.B \simeq C.A.B$; ergo divellendo et $F.A \simeq C.A$, itemque $F.B \simeq C.B$; ergo F cadet in ambas circumferentias, atque
 10 adeo vel coincidit cum C , vel C non erit solum commune, quod est absurdum.

in the same line as the centres. Let the centres be A and B , and the point of contact C , namely where two meeting points merge. And so (by the preceding) the circle does not meet the circle elsewhere, otherwise there will be three meeting points. Therefore only the point of contact is common to the two circles. I claim it is in the line through A, B . This is apparent from §19, if it is shown that it is unique of its relation to A, B . Let there be another one F , if this can happen, and one should have $F.A.B \simeq C.A.B$. Therefore by disjoining, both $F.A \simeq C.A$ and also $F.B \simeq C.B$. Therefore F falls in both circumferences, and thus either they coincide with C , or C is not the only thing in common, which is absurd.

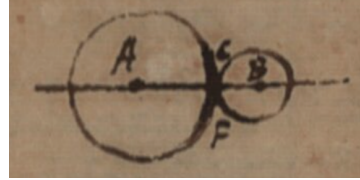


Figure 11

(32) Recta et circulus non possunt sibi occurrere in plus quam duobus punctis. Sint L et M (fig. 76) in recta $\bar{X}$, itemque in circumferentia $\bar{Z}$ circa C , dico praeter L et M non posse dari punctum N . Sumatur D eodem modo se habens ad rectam $\bar{X}$,
 15 ut C ab altera parte per § 17. Ob circulum est $L.C \simeq M.C \simeq N.C$, ergo quia puncta L, M, N sunt in recta eodem modo se habente ad D , quo ad C , etiam erit $L.D \simeq M.D \simeq N.D$; ergo conjungendo $L.C.D \simeq M.C.D \simeq N.C.D$. Sit $\bar{Y}$ recta per $C.D$,
 20 ergo (per axiom. § 3.) fiet $L.\bar{Y} \simeq M.\bar{Y} \simeq N.\bar{Y}$, seu recta $\bar{Y}$ se eodem modo habebit ad tria puncta L, M, N , quod fieri nequit per § 29.

(32) A line and a circle cannot meet each other in more than two points. Let L and M be in line $\bar{X}$ [Figure 12], and also in circumference $\bar{Z}$ around C . I say that besides L and M , another point N could not be given. Let D be taken relating in the same way to the line $\bar{X}$ as C on the other side, by §17. On account of the circle, $L.C \simeq M.C \simeq N.C$ holds. Therefore since the points L, M, N are in a line, they relate in the same way to D as to C , and also $L.D \simeq M.D \simeq N.D$. Therefore by combining, $L.C.D \simeq M.C.D \simeq N.C.D$. Let $\bar{Y}$ be the line through $C.D$; then (by the axiom of §3) $L.\bar{Y} \simeq M.\bar{Y} \simeq N.\bar{Y}$ will hold, i.e. the line $\bar{Y}$ will relate in the same way to the three points L, M, N , which cannot happen by §29.

Atque ita fundamentalia rectae et circuli exposuimus, quomodo scilicet occurrere sibi possint haec loca: recta rectae,
 25 circulus circulo, recta circulo, quorum occursibus caetera determinantur. Unde consequens est, caetera quoque calculo

And so we have revealed the foundations of the straight line and the circle, namely the way these loci can meet each other: a line and a line, a circle and a circle, a line and a circle, by the meetings of which others are determined. From this it follows that the others can also

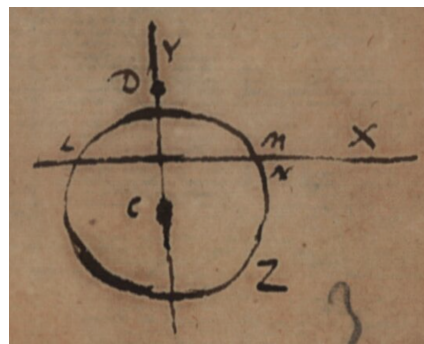


Figure 12

nostro tractari posse.

be handled by our calculus.